← Back
Practice Quiz: Introduction to Web Tech Web
Practice Assignment, 20 min.
Due: Aug 27, 2:59 PM IST

Your grade: 100%

You scored **100%**. The highest **100%** is. To pass you need at least 80%. Review your highest score.

Retake Quiz

Attempt number this assignment: **1**

Advanced KAG System Implementation	1 / 1 pts Graded on

---

1. What are the differences between locally-weighted hashing (LSH) and Minwise Independent Permutations (MIP) algorithms? 1 / 1 point

- MIPM is a LSH that uses 1 whereas LSH uses distributed hashing.
- LSH is used for approximate nearest neighbor search hashing functionalities, whereas MIPM is a graph-based index that organizes our terms by the PLURALITY of their search.
- LSH provides exact search results, whereas MIPM is used for hashing vectors.
- Both LSH and MIPM are used for exact search in NLP.

Your work

LSH hashes similar vectors (terms/documents) for fast lookup, whereas MIPM uses 1 (hashed) graph data (term lists affiliated to each).

2. How do NGS and Chromium differ in their indexing methods and capabilities? 1 / 1 point

- NGS and Chromium differ in their indexing capabilities, resulting in their interoperability.
- Chromium supports hierarchical storage and search indexing, while NGS.
- Both NGS and Chromium support parallel hashing and fast process indexing methods.
- NGS supports multiple index types such as HC, PLA, and MIGRE, whereas Chromium (Blueprinted) supports MIPM & HC in its storage.

Your work

Both offer a variety of indexing strategies, whereas Chromium differentiates in MIPM.

3. Under what conditions is MIPM a better choice than Chromium (MIPM) for vector similarity search? 1 / 1 point

- NGS is preferable only when matching with minhash only.
- NGS is preferable when storage vector storage additional storage (indexing) is needed.
- NGS is always preferable due to its simplicity and ease of use.
- NGS is preferable when high performance indexing and efficient similarity search are required in a single machine.

Your work

NGS is ideal for advanced, local-scale search tasks.

4. What is the purpose of the hierarchical structure in the MIPM algorithm? 1 / 1 point

- It allows the algorithm to skip the vector generation step by using random walks.
- It enables fast and scalable search by combining large-scale and short-range connections across layers.
- It increases the indexing speed as it reduces the number of the Jaccard score memory usage.
- It guarantees exact nearest neighbor results by using binary trees to search each level.

Your work

The top layers provide fast traversal, across the data space, while the bottom layers offer fine-grained accuracy.

5. Which of the following is a known trade-off when using the MIPM algorithm? 1 / 1 point

- MIPM always returns the exact nearest neighbor, regardless of parameter settings.
- Using fewer layers improves both speed and accuracy.
- The MIPM algorithm performs best with frequent insertions and deletions in dynamic datasets.
- Higher accuracy requires more memory and longer build times due to increased graph complexity.

Your work

Parameters can both affect efficiency and improve accuracy, but increase memory and build time.